Computer Science & IT

COMPILER DESIGN
(CD)

Parsers

Lecture No. 2



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Grammar

Topics to be Covered



- Topic
- Topic
- Topic

Grammar

Panch (90%)



Unstoppable Indians: Stories to Ignite Student Motivation

Edavaleth Kakkat Janaki Ammal (1897–1984) was a pioneering botanist who co-discovered colchicine-induced polyploidy, enhancing crop varieties. Educated in the UK, she returned to lead genetic and cytogenetic research at the Sugarcane Breeding Institute and the Botanical Survey of India. Ammal championed biodiversity conservation and mentored women scientists, promoting indigenous plant research to improve food security.

Lesson: Scientific innovation and mentorship can advance agriculture and empower underrepresented communities.



Unstoppable Indians: Stories to Ignite Student Motivation

Asima Chatterjee (1917–2006) was India's first female PhD in science, renowned for her work in organic chemistry and pharmacology. She discovered anti-epileptic and anti-malarial compounds from indigenous plants and developed vinca alkaloids for cancer treatment. Leading the University of Calcutta's chemistry department, she fostered a research culture and authored influential textbooks.

Lesson: Research driven by societal needs can yield life-saving treatments and inspire future generations.



Unstoppable Indians: Stories to Ignite Student Motivation

Kamala Sohonie (1912–1998) broke barriers as India's first woman to earn a PhD in science. Studying under C. V. Raman, she researched plant protein metabolism at the Indian Institute of Science. Despite gender biases, Sohonie tirelessly published her findings and taught biochemistry, laying groundwork for food science in India and paving the way for women scientists.

Lesson: Perseverance against prejudice can open doors for inclusive scientific progress.



Unstoppable Indians: Stories to Ignite Student Motivation

Koccharlakota Suryanarayana Krishnan (1908–1961) co-discovered the Raman scattering effect alongside C. V. Raman. Later, he conducted pioneering work in magnetism and electron spin resonance. As Professor at Allahabad University, he mentored young physicists and established research laboratories, strengthening India's foundational physics programs.

Lesson: Collaborative discovery and dedication to mentorship foster scientific communities that thrive across generations.

$E \rightarrow E + T / T$

$T \rightarrow T * F / F$

$F \rightarrow G \uparrow F / G$

$G \rightarrow id$

$+ \gtrdot +$

$* \gtrdot *$

$\uparrow \lessdot \uparrow$

$id \gtrdot \uparrow \gtrdot * \gtrdot +$

disambiguity Rules

$LR \rightarrow LA$

$RR \rightarrow RA$

level 1 → least precem

level n → high precedence

Boolean Expressions :-

bExpr → bExp or bExp
/ bExp and bExp
/ not bExpr
/ ( )
/ True
/ False

S → (S)

OR - LA - LR
and - LA - LR
not -
true
false

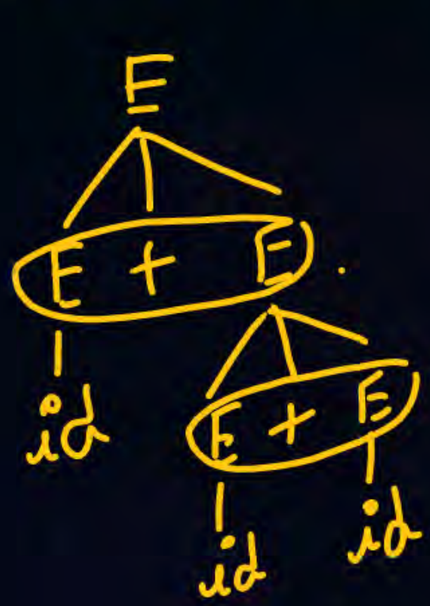
E → E or F / F
F → F and G / G
G → not G / True / False

or > or
and > and

or ≤ and ≤ not

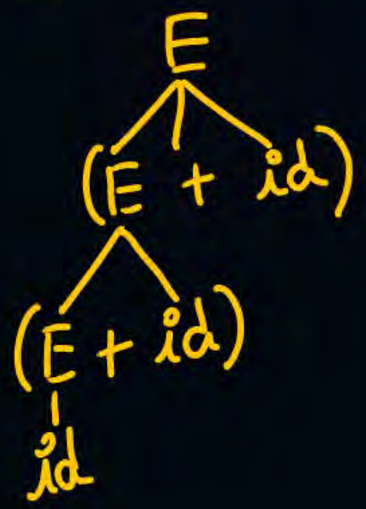
$E \rightarrow E+E/id$

$id+id+id$

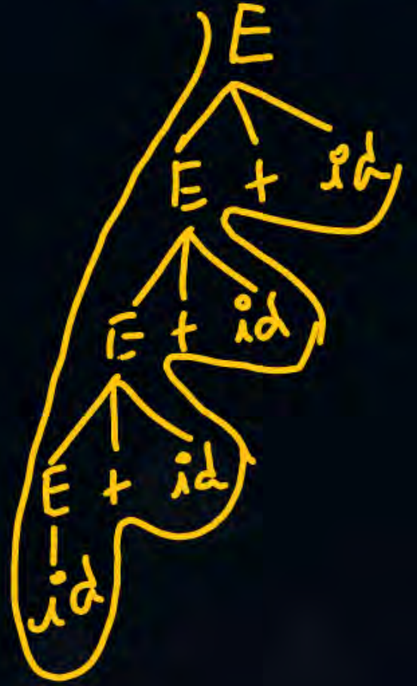




$E \rightarrow E+id/id$



$id+id+id+id$



If the production is left recursive then the operator becomes left ass

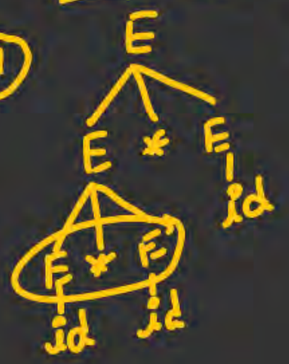
$E \rightarrow E * E / id$

$E \rightarrow E * id / id$

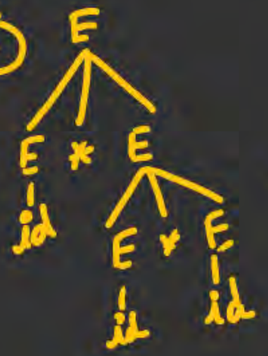
id * id * id

id * id * id * id

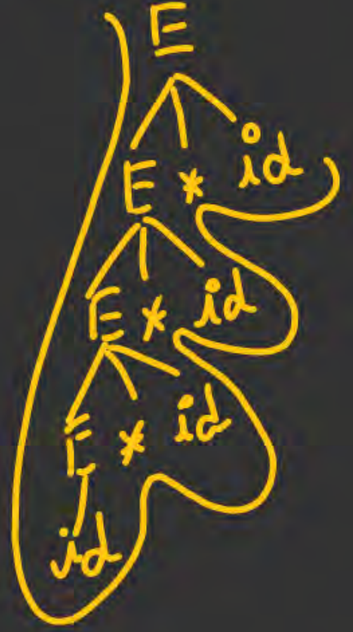
(1)



(2)







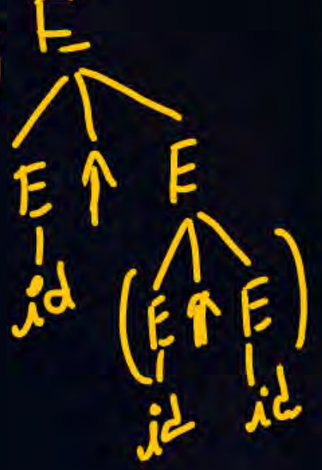
$(2)^{(3^4)} \Rightarrow (2 \uparrow (3 \uparrow 4))$

$E \to E \uparrow E / id$

$id \uparrow (id \uparrow id)$



(2)



$E \to id \uparrow E / id$

$* > +$

$E \to E + T / T$ ✓

$T \to T * F / F$ ✓

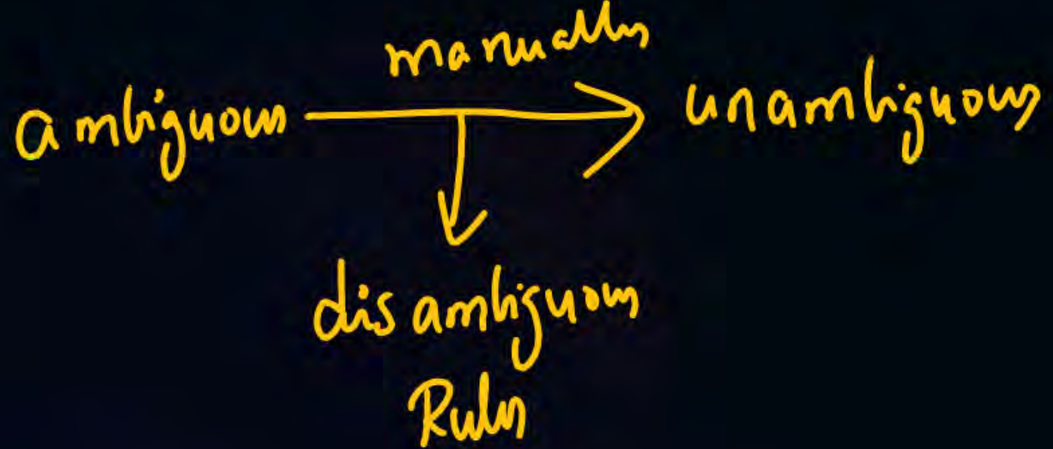
$F \to id$

$id * id + id$



→ Checking whether a grammar is ambiguous is undecidable → no algo

→ Converting an ambiguous grammar to ambiguous grammar is undecidable
↓
NO



Grammar for Regular expressions :

$R \to R+R$
$/RR$
$/R^*$
$/a$
$/b$
$/\epsilon$



$E \to E+T/T$

$T \to TF/F$

$F \to F^*/a/b/\epsilon$

Gate - 2015

$A \rightarrow \underline{A}\ \$\ B \mid B$

$B \rightarrow \underline{B}\ \#\ C \mid C$

$C \rightarrow \underline{C}\ @\ D \mid D$

$D \rightarrow \underline{d}$

$\$ \gtrdot \$$

$\# \gtrdot \#$

$@ \gtrdot @$

$\$ \lessdot \# \lessdot @ \lessdot d$

ambo, prec

Gate 2021 :- Is the grammar ambiguous

$E \to E * F / F + E / F$

$F \to F - F / id$

$* = +$

$id + (id * id)$ ✓

$(id + id) * id$ ✓

$* \doteq +$

$* \gtrdot *$

$+ \lessdot +$

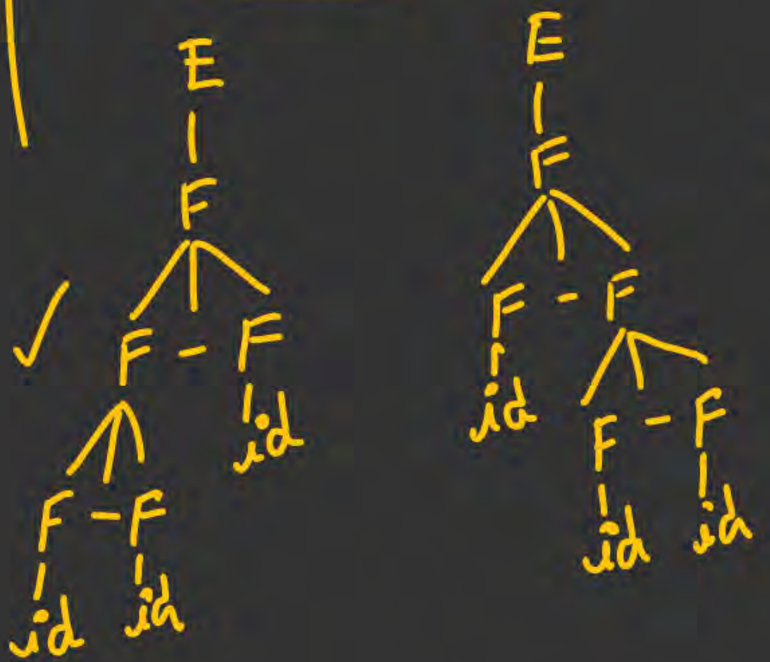
$- \gtrdot * \doteq +$

$-$ → ans is not defined

$E \to \underline{E * F} / \underline{F + E} / F$

$F \to \underline{F - F}$

$\underline{id - id - id}$



ambg

$E \rightarrow E + E$
$\quad / \; E * E$
$\quad / \; id$

$+ < *$

$E \rightarrow E+E / E*E / id$

$+ = * = id$

Grammar

- **Ambi**
  - no parse can parse ambiguous grammar (except OPP)
- **Unambi**
  - All unambi grammar cannot be parsed by any parser

Grammar

- **Left Recursive**
  - Top down parser cannot parse LR
- **Right recursive**
  - all Right recursive grammar cannot be parsed by all parser

Grammar

- **deterministic**
  - all deterministic grammar cannot be parsed by any parser
- **Non deterministic**
  - Top down parser cannot parse non deter grammar

TDP Cannot Work with Left recursive

Grammar

Left Recursive

Right recursive ✓

$A \to A\alpha / \beta$

$A \to \alpha A / \beta$



$\beta \alpha^*$

$A \to \beta A'$

$A' \to \alpha A' / \epsilon$

Right Recursive grammar

X



$\alpha^* \beta$

Gate 2014:

$E \to E+T/T$

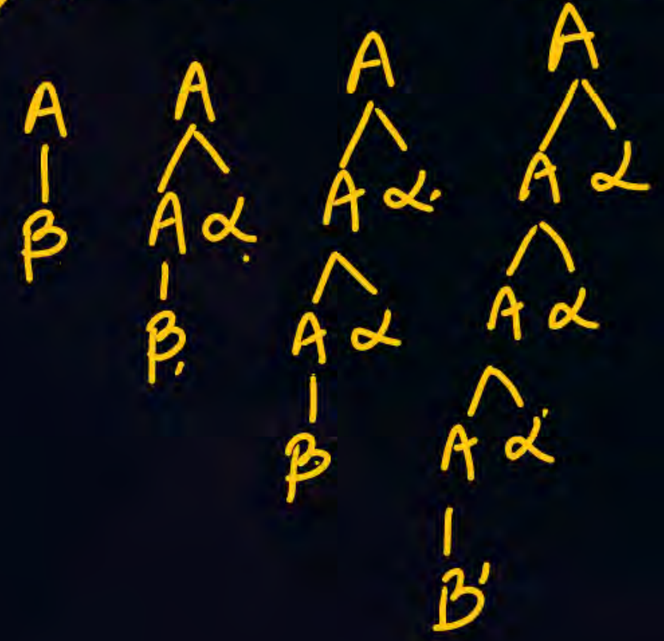
$A \to A\alpha/\beta$

$A \to \beta A'$

$A' \to \alpha A'/\epsilon$

$A \to A\alpha/\beta$ (LLG) $\Longrightarrow$ $\beta\alpha^*$

RRG



$A \to \beta A'$

$A' \to \alpha A'/\epsilon$

Right Recur

$E \to E+T / T$

$A \quad A\alpha \quad \beta$

$A \to \beta A'$

$A' \to \alpha A' / \epsilon$

$E \to TE'$

$E' \to +TE' / \epsilon$

Grati:-

$S \to S0S1S / 01$

$A \quad A \quad \alpha \quad \beta$

$A \to \beta A'$

$A' \to \alpha A' / \epsilon$

$S \to 01S'$

$S' \to \alpha S' / \epsilon$

Gate:-

$S \rightarrow (L) / x$ ✓

$L \rightarrow L, S / S$ ✓

$L \rightarrow L, S / S$

$A \quad A \alpha \quad \beta$

$A \rightarrow \beta A'$

$A' \rightarrow \alpha A' / \epsilon$

$L \rightarrow S L'$

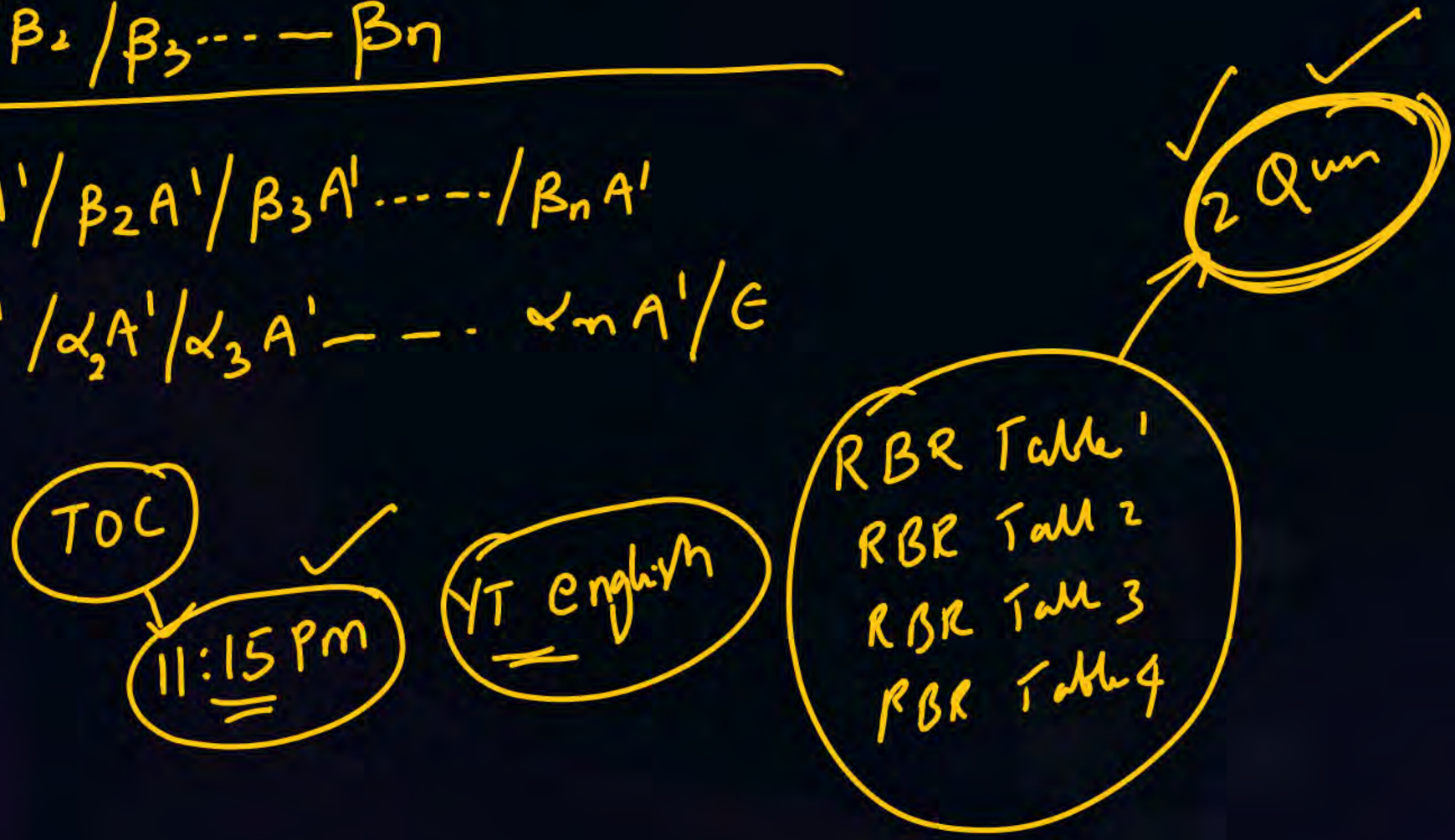
$L' \rightarrow , S A' / \epsilon$

$A \rightarrow A\alpha_1 / A\alpha_2 / A\alpha_3 \cdots A\alpha_n$

$/\beta_1 / \beta_2 / \beta_3 \cdots \beta_n$

$A \rightarrow \beta_1 A' / \beta_2 A' / \beta_3 A' \cdots / \beta_n A'$

$A' \rightarrow \alpha_1 A' / \alpha_2 A' / \alpha_3 A' \cdots \alpha_m A' / \epsilon$



THANK - YOU